

Paper_ED_Cos_02 — Big Bang Nucleosynthesis as Substrate-Inherited Freeze-Out Sequence

Series: Wave-3 Generative Papers (Cosmology Arc — early-universe nucleosynthesis) **Status:** Verdict per Paper_095: **M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED)** within DCGT A→regime. Direct composition of Cos_01 + Cos_05 expansion-history framework + Route A4 substrate-parameter inheritance + standard nuclear physics + standard BBN-code machinery via DCGT continuum bridge. **M3 via continuum-Friedmann inheritance** — Dyn_02’s M2 RDE chain-class identification is *not* load-bearing for Cos_02 (see Preamble item 7). **Date:** 2026-05-17 **Authors:** Paper_087 (13 primitives); Paper_073 (DCGT); Paper_012 (substrate-c); Paper_027 (Newton’s G); Paper_028 ($R_H = c/H_0$); Paper_ED_SC_4_9 (S1/S2/S3); Paper_ED_Cos_01 (M3 — Inflation); Paper_ED_Cos_05 (M3 — Dark Energy); Paper_ED_Dyn_02 (M2 — RDE, upstream-status only); Paper_095 (verdict grammar — §3.2.1 sub-labels). Substrate-research support memos cross-referenced in §2.

Label-class note: Audit-row sub-labels *substrate-parameter-INHERITED* and *IC-INHERITED* per Paper_095 §3.2.1.

Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim closed-proof derivation of standard nuclear-physics machinery from substrate primitives. Standard nuclear physics (weak-interaction rates; nuclear reaction cross-sections; deuterium binding energy; etc.) is **INHERITED**; ED supplies the substrate-graph reading of *what the radiation-era $H(t)$ is and how it enters the nuclear-freeze-out competition* — not a substrate-graph rederivation of nuclear physics.
2. It does **not** sit within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). ED is in the substrate-ontology research program. Methodology per Paper_095 (form-FORCED / value-INHERITED).
3. It does **not** introduce new substrate primitives or paper-specific postulates. The BBN content is composition of Cos_01 + Cos_05 + Route A4 substrate-parameter-INHERITED scale + standard nuclear physics + standard cosmological-Friedmann RDE framework via DCGT continuum bridge.
4. It does **not** claim numerical-precision derivation of specific BBN abundance ratios (Y_p 0.247; D/H 2.5×10^{-5} ; Li/H 1.5×10^{-10}). Specific numerical values inherit from standard BBN-code calculations (e.g., PArthENoPE, AlterBBN) executed under the standard cosmological + nuclear-physics framework; ED supplies the substrate-graph mechanism that *produces* the standard $H(t)$ used in those calculations, not a substrate-graph rederivation of all numerical abundance values.
5. It does **not** address the “primordial lithium problem” (observed Li/H lower than standard-BBN prediction by factor 3-4). This is a standard-BBN puzzle independent of substrate-ontology framing; ED inherits the standard-BBN Li prediction at standard-cosmology level and does not supply a substrate-graph mechanism specifically resolving the puzzle. Standard-physics resolutions (e.g., stellar depletion; new-physics scenarios) are inheritance-level options also available to ED.
6. It does **not** address pre-BBN cosmological-history details (specific reheating mechanism; particle-physics content prior to weak freeze-out) beyond inheriting Cos_01 post-inflation reheating framework + standard cosmology.

7. **Inheritance from Dyn_02 M2 status (architectural division).** Cos_02 inherits standard-cosmology continuum-Friedmann $H \propto T^2$ via DCGT continuum bridge at M3-equivalent inheritance level. Dyn_02's substrate-graph RDE chain-class identification (M2 with OPEN-HM-Q1 + OPEN-HM-Q2: chain-class criterion + stress-energy form per regime) remains OPEN at Dyn_02's level; **Cos_02 does not load-bear on these closures** because BBN's substrate-graph content (Route A4 $H_0 + \Theta_{\text{ED}}$ sensitivity channel) is independent of the chain-class identification for RDE matter content. Same architectural-division logic as Dyn_01 + Dyn_04 + GW_01 (foundational / continuum-inheritance pathway vs substrate-graph chain-class pathway).
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Abstract

Big Bang Nucleosynthesis (BBN) is identified at substrate-graph level as **the inherited competition between standard nuclear reaction rates Γ_{nuc} and the substrate-graph-derived expansion rate $H(t)$ during the radiation-dominated era.** $H(t)$ inherits continuum-side Friedmann $H \propto T^2$ via DCGT continuum bridge at M3-equivalent level (Cos_01 + Cos_05 supply the cosmological-expansion framework); substrate-parameter H_0 enters at substrate-parameter-INHERITED level via Route A4 (Paper_027-analog); standard nuclear physics + standard BBN-code calculations inherit at textbook level. Three temporal regimes: weak freeze-out at $T \sim 1$ MeV $\rightarrow$ deuterium bottleneck at $T \sim 0.07$ MeV $\rightarrow$ helium-4 synthesis ($Y_p \approx 0.247$). **Form-IDENTIFIED** via Cos_01 + Cos_05 + DCGT + standard cosmology + standard nuclear physics. **Value-INHERITED:** abundance ratios via standard BBN codes; H_0 via Route A4 substrate-parameter inheritance. **IC-INHERITED** from Cos_01 post-inflation reheating + CMB-derived η_b . Substrate-c preserved. **Verdict per Paper_095: M3** (form-IDENTIFIED + value-INHERITED + IC-INHERITED); Dyn_02 M2 status *not* load-bearing (Preamble item 7). **F1 (sharpest):** Y_p or D/H significantly inconsistent with substrate-graph $H(t)$ + standard nuclear physics — beyond $\sim 5\%$ standard-BBN uncertainty — refutes the composition. **Empirical anchors are exceptionally strong:** 50+ years of BBN observational success at $\sim 5\%$ precision; $Y_p^{\text{obs}} \approx 0.245\text{--}0.255$; $\text{D}/\text{H}^{\text{obs}} \approx (2.5\text{--}2.6) \times 10^{-5}$; CMB-BBN η_b cross-consistency at $\sim 5\%$ (Planck $\eta_b \approx 6.1 \times 10^{-10}$). The substrate-parameter Θ_{ED} sensitivity channel ($dY_p/d \ln \Theta_{\text{ED}} \approx 0.05$) is the genuine ED-distinguishing content. Enables Paper_ED_Cos_03 (CMB acoustic) + Paper_ED_Cos_04 (linear structure formation).

§1 Statement of Result

BBN is the substrate-graph **freeze-out competition** between nuclear-reaction rates Γ_{nuc} (textbook nuclear physics) and the radiation-era expansion rate $H(t)$ (inherits standard cosmology + Route A4 substrate-parameter-INHERITED scale).

Three temporal regimes (standard BBN phenomenology, substrate-side reading):

Regime	T scale	Time	Substrate-graph content
Weak freeze-out	$T \sim 1$ MeV	$t \sim 1$ s	Weak-interaction rate $\Gamma_W \sim G_F^2 T^5$ falls below $H(t) \propto T^2$; n/p ratio freezes at $\sim 1/6$, drops to $\sim 1/7$ via neutron decay
Deuterium bottleneck	$T \sim 0.07$ MeV	$t \sim 3$ min	D photodissociation rate $\propto e^{-B_D/T}$ ($B_D = 2.22$ MeV) falls below D-formation rate; nuclei begin forming
Helium synthesis	$T \sim 0.05$ MeV	$t \sim 10$ min	Bulk of free neutrons $\rightarrow$ He-4 via $p + n \rightarrow D + \gamma$, $D + D \rightarrow T + p$ (or ${}^3\text{He} + n$), etc.; $Y_p \approx 0.247$; D/H 2.5×10^{-5} ; trace ${}^3\text{He}$, ${}^7\text{Li}$

FORM: BBN = freeze-out competition between Γ_{nuc} and substrate-graph-derived $H(t)$ — **IDENTIFIED** via Cos_01 + Cos_05 + Route A4 + standard cosmology + standard nuclear physics composition. Zero pure-D rows.

VALUE: H_0 at substrate-parameter-INHERITED level via Route A4 (Paper_027-analog); RDE $H(t) \propto T^2$ inherits standard cosmology; specific abundance ratios (Y_p , D/H, etc.) inherit standard-BBN code calculations.

IC: early-universe substrate state (radiation-dominated era post-inflation; baryon-to-photon ratio η_b ; particle content) inherited from standard cosmology + Cos_01 inflation framework + CMB-derived η_b .

Verdict: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A $\rightarrow$ regime per §4 audit table.

§2 Primitive Inputs + Upstream Dependencies

Primitives (Paper_087): P02 (participation), P03 (translation invariance), P09 (polarity), P10 (multiple rule-types), P11 (commitment / time arrow), P12 (ED-threshold + stability landscape), P13 (translation arrow).

Upstream papers: - Paper_012 (P-RB-1 substrate-c bound) - Paper_027 (Newton's G — substrate-parameter inheritance for Friedmann G) - Paper_028 (cosmic decoupling surface R_H — cosmological-history framework) - Paper_073 (DCGT hydrodynamic-window coarse-graining)

— supplies continuum-side Friedmann inheritance at RDE) - Paper_ED_SC_4_9 (saddle classification S1/S2/S3) - Paper_ED_Cos_01 (Inflation, M3 — supplies post-BBB cosmological initial conditions + post-inflation reheating framework + early-universe substrate state) - Paper_ED_Cos_05 (Dark Energy, M3 — supplies late-universe expansion-history asymptote + Route A4 substrate-parameter inheritance for H_0) - Paper_ED_Dyn_02 (Horizon-Motion, M2 — supplies substrate-graph RDE regime identification at upstream M2 status; Cos_02 inherits this at upstream M2 status as substrate-research-frontier tightening, not load-bearing) - Update_ED_SC_4x_Arc_M3 (six-projection arc-wide M3 upgrade post-Route-A) - Standard cosmological Friedmann machinery at RDE ($H \propto T^2$; $T \propto a^{-1}$) — textbook inheritance via DCGT continuum bridge - Standard nuclear-physics machinery (weak-interaction rates $\Gamma_W \sim G_F^2 T^5$; nuclear reaction cross-sections; deuterium binding energy $B_D = 2.22$ MeV; standard particle content) — textbook inheritance - Standard BBN-code calculations (PARthENoPE, AlterBBN, etc.) — textbook inheritance for specific numerical abundance predictions

Substrate-research support: - Memo_ED_RouteA_A4_Audit + Memo_ED_RouteA_MultiRouteConvergence (Route A4 substrate-parameter-INHERITED closure at Paper_027-analog tier; supplies H_0 scaling) - Memo_ED_Q1Q2_JointClosure_Construct + Memo_ED_ChainClass_C3_Construct + Audit (saturation chain framework; non-saturation RDE chain-class identification is Dyn_02 M2 territory)

No new substrate primitives. No paper-specific postulates.

§3 Derivation

§3.1 ED expansion rate $H(t)$ in radiation era

During the radiation-dominated era (RDE; from end of inflation/reheating until matter-radiation equality $z_{\text{eq}} \approx 3400$), the cosmological-history expansion rate is

$$H(t) = \sqrt{\frac{8\pi G}{3} \rho_{\text{rad}}}, \quad \rho_{\text{rad}} = \frac{\pi^2}{30} g_*(T) T^4$$

giving $H \propto T^2$ at RDE. **ED inherits this continuum-side Friedmann content at standard-cosmology level via DCGT continuum bridge** (Paper_073). The substrate-graph identification of *why* the substrate behaves as $w = 1/3$ during RDE is Paper_ED_Dyn_02 M2 territory (Route C1 substrate-graph parameter constructions pending); Cos_02 inherits Dyn_02's continuum-side $w = 1/3$ identification at standard-cosmology level without depending on the substrate-graph chain-class identification for Cos_02's net verdict.

Substrate-parameter content: G via Paper_027 dimensional rearrangement; H_0 via Route A4 substrate-parameter-INHERITED chain (Paper_027-analog per Memo_ED_RouteA_MultiRouteConvergence_A option (ii)): $H_0 = (8\pi/3)^{1/2} \ell_P c^{1/2} \Theta_{\text{ED}}^{1/2}$ with $\Theta_{\text{ED}}^{\text{Planck-units}} \approx 10^{-122}$ at primitive level. The overall cosmological-history time-scale calibration enters BBN through H_0 + standard cosmology evolution to BBN epoch.

§3.2 Weak freeze-out (n/p ratio)

The weak-interaction rate maintaining $n \leftrightarrow p$ equilibrium scales as $\Gamma_W \sim G_F^2 T^5$ (standard particle physics; Fermi constant G_F). Freeze-out occurs when $\Gamma_W \sim H(t)$:

$$T_{\text{freeze}} \sim \left(\frac{H(t)}{G_F^2} \right)^{1/3} \approx 1 \text{ MeV}.$$

At T_{freeze} , the n/p equilibrium ratio is $\exp(-\Delta m/T_{\text{freeze}}) \approx 1/6$ (with $\Delta m = m_n - m_p \approx 1.29 \text{ MeV}$). After freeze-out, neutron decay ($\tau_n \approx 880 \text{ s}$) reduces the n/p ratio to $\sim 1/7$ by the time deuterium formation begins.

Substrate-graph composition: standard particle physics Γ_W + substrate-graph-inherited $H(t)$. The T_{freeze} value inherits standard cosmology + Route A4 substrate-parameter-INHERITED H_0 scaling at the BBN epoch.

§3.3 Deuterium bottleneck

The deuterium photodissociation rate dominates over D-formation while the radiation field maintains substantial γ -flux above the D binding energy $B_D = 2.22 \text{ MeV}$. Photodissociation rate scales as $\Gamma_{\gamma D} \propto \eta_b T^{3/2} \exp(-B_D/T)$; D-formation rate via $p + n \rightarrow D + \gamma$. Bottleneck breaks when photodissociation rate falls below D-formation rate:

$$T_{\text{D-bottle}} \approx \frac{B_D}{\ln(\eta_b^{-1}) - \frac{3}{2} \ln T} \approx 0.07 \text{ MeV}.$$

At $T_{\text{D-bottle}}$, D begins to accumulate; nuclear reactions $D + D \rightarrow T + p$, $D + D \rightarrow {}^3\text{He} + n$, $D + T \rightarrow {}^4\text{He} + n$, etc., initiate. Substrate-graph composition: standard nuclear physics + baryon-to-photon ratio η_b inherited from CMB observations + substrate-graph-inherited $H(t)$ controlling the time-scale for nuclear-reaction sequence to complete.

§3.4 Helium-4 synthesis

Once the D bottleneck breaks, nearly all available free neutrons are captured into He-4 via the chain $p + n \rightarrow D \rightarrow D + D \rightarrow {}^3\text{He} + n$ or $D + D \rightarrow T + p \rightarrow {}^3\text{He} + D \rightarrow {}^4\text{He} + p$ (and analogous T-mediated paths). The He-4 mass fraction:

$$Y_p = \frac{2(n/p)_{\text{D-bottle}}}{1 + (n/p)_{\text{D-bottle}}} \approx 0.247$$

with $(n/p)_{\text{D-bottle}} \approx 1/7$ inherited from §3.2 + §3.3. Substrate-graph composition: standard nuclear-physics neutron-capture chain + substrate-graph-inherited $H(t)$ + standard cosmology time-scale.

The substrate-graph-inherited $H(t)$ sensitivity: faster expansion (larger H) shortens weak-freeze-out time $\rightarrow$ less neutron decay before D-bottle $\rightarrow$ higher $(n/p)_{\text{D-bottle}} \rightarrow$ higher Y_p . Quantitative sensitivity $dY_p/d(\delta H/H) \approx 0.1$ at standard expansion rate (textbook BBN sensitivity).

§3.5 Light-element abundances

Standard BBN predictions (PArthENoPE / AlterBBN computational frameworks applied under the standard cosmological + nuclear-physics framework, with $H(t)$ inherited from §3.1):

- $Y_p \approx 0.247$ (He-4 mass fraction)

- $D/H \approx 2.5 \times 10^{-5}$
- ${}^3\text{He}/H \approx 1.0 \times 10^{-5}$
- ${}^7\text{Li}/H \approx 1.5 \times 10^{-10}$

These specific numerical predictions inherit from standard-BBN-code calculations + standard nuclear physics + standard cosmological framework + substrate-parameter-INHERITED H_0 via Route A4. ED supplies the substrate-graph mechanism that *produces* the standard $H(t)$ used in those calculations; ED does not supply a substrate-graph rederivation of the specific abundance values beyond inheriting standard BBN-code outputs.

Observational consistency: $Y_p^{\text{obs}} \approx 0.245\text{--}0.255$ (Izotov-Thuan-style metal-poor extragalactic HII regions; Aver et al.); $D/H^{\text{obs}} \approx (2.5\text{--}2.6) \times 10^{-5}$ (high- z QSO absorption lines; Cooke et al.); ${}^7\text{Li}/H^{\text{obs}} \approx (1\text{--}2) \times 10^{-10}$ at metal-poor stars (Spite plateau; below standard-BBN prediction — the “lithium problem”). $\eta_b^{\text{CMB}} \approx 6.1 \times 10^{-10}$ from Planck consistent with η_b^{BBN} at $\sim 5\%$ precision (modulo lithium puzzle).

§3.6 Sensitivity to Θ_{ED} or τ_{V_5}

The substrate-parameter content Θ_{ED} enters BBN through Route A4’s identification $H_0 \propto \Theta_{\text{ED}}^{1/2}$. Standard cosmology evolution from BBN epoch to present supplies $H(z_{\text{BBN}})/H_0$ at large redshift. The overall BBN time-scale and freeze-out temperatures inherit the substrate-parameter scaling:

- Larger $\Theta_{\text{ED}} \rightarrow$ larger $H_0 \rightarrow$ larger $H(t)$ at all epochs (proportionally) $\rightarrow$ earlier weak freeze-out at higher $T_{\text{freeze}} \rightarrow$ higher $(n/p)_{\text{freeze}} \rightarrow$ higher Y_p
- Quantitative sensitivity: $dY_p/d \ln \Theta_{\text{ED}} \approx 0.05$ at observed substrate-parameter value, composed as Route A4 $H_0 \propto \Theta_{\text{ED}}^{1/2}$ (so $d \ln H_0/d \ln \Theta_{\text{ED}} = 1/2$) $\rightarrow$ standard BBN H -sensitivity $dY_p/d(\delta H/H) \approx 0.1$ (textbook; e.g. Steigman 2007) $\rightarrow dY_p/d \ln \Theta_{\text{ED}} \approx 0.5 \times 0.1 \approx 0.05$. This is a composition of Route A4 substrate-parameter inheritance + standard BBN sensitivity; honest D-via-I, not an independent assertion.

Tightening via RA-OPEN-4a (substrate-graph derivation of Θ_{ED} value from sub-primitive content) would refine BBN predictions correspondingly; substrate-research-frontier, not load-bearing for Cos_02’s M3 verdict.

Similarly τ_{V_5} enters via Route A1/A2 alternative (multi-route convergence option (ii)) at same substrate-parameter-INHERITED level.

§3.7 Comparison to Λ CDM BBN

Element	Λ CDM standard BBN	ED substrate-graph reading	Status
Expansion rate $H(t)$ at RDE	Continuum Friedmann $H \propto T^2$ on FRW background	DCGT-inherited continuum Friedmann at standard-cosmology level	Inherited (substrate-graph supplies what underlies the inheritance)

Element	Λ CDM standard BBN	ED substrate-graph reading	Status
Substrate-graph identification of RDE	None (continuum-only ontology)	Mixed-signature non-saturation substrate state per Dyn_02 (M2 — Route C1 pending)	Distinguishing content (substrate-graph identification supplied at Dyn_02 upstream M2; not load-bearing for Cos_02 net M3)
Weak freeze-out, D bottleneck, He-4 synthesis sequence	Standard nuclear physics	Same standard nuclear physics + substrate-graph $H(t)$ inheritance	Inherited + Identified
Specific abundance predictions (Y_p , D/H, etc.)	Standard BBN codes	Same standard BBN codes; substrate-parameter sensitivity via Route A4	Inherited
Baryon-to-photon ratio η_b	Free parameter constrained by CMB + BBN cross-consistency	Same; no substrate-graph rederivation of η_b	Inherited
Lithium problem (observed below prediction)	Unresolved standard-BBN puzzle	Same; ED inherits puzzle, does not specifically resolve	Inherited puzzle
Substrate-parameter dependence	None	$H_0 \propto \Theta_{\text{ED}}^{1/2}$ via Route A4 $\rightarrow$ BBN abundances inherit substrate-parameter sensitivity	Distinguishing content (substrate-parameter scaling channel)

§4 Audit Table

#	Step	Label	Notes
1	P02, P03, P09, P10, P11, P12, P13	P	Paper_087.

#	Step	Label	Notes
2	Upstream inheritance block	I	Paper_012; Paper_027; Paper_028; Paper_073; Paper_ED_SC_4_9; Paper_ED_Cos_01 (M3); Paper_ED_Cos_05 (M3); Paper_ED_Dyn_02 (M2 — upstream, not load-bearing for Cos_02 net M3 via continuum-Friedmann inheritance) ; Update_ED_SC_4x_Arc_M3.
3	Standard cosmological Friedmann RDE machinery ($H \propto T^2$; $T \propto a^{-1}$)	I	Textbook inheritance via DCGT continuum bridge.
4	Standard nuclear physics (G_F weak-interaction rates; nuclear cross-sections; $B_D = 2.22$ MeV)	I	Textbook inheritance.
5	Standard BBN-code calculations (PArthENoPE / AlterBBN)	I	Textbook inheritance for specific numerical abundances.
6	$H(t)$ at RDE via continuum-Friedmann inheritance (substrate-graph RDE identification at upstream Dyn_02 M2; net M3 inheritance via DCGT)	D-via-I	§3.1. Cos_01 + Cos_05 + DCGT + Paper_027 + Route A4 composition.

#	Step	Label	Notes
7	H_0 at substrate-parameter- INHERITED level via Route A4	I	Sub-type per Paper_095 §3.2.1: <i>substrate-parameter- INHERITED</i> . §3.1, §3.6. Paper_027-analog tier per Memo_RouteA_MultiRouteConverge option (ii).
8	Weak freeze-out at $T_{\text{freeze}} \sim 1$ MeV from $\Gamma_W \sim H(t)$ competition	D-via-I	§3.2. Standard Γ_W + substrate-graph- inherited $H(t)$.
9	n/p ratio at freeze-out $\sim 1/6$, drops to $\sim 1/7$ via neutron decay	D-via-I	§3.2. Standard equilibrium + neutron-decay inheritance.
10	Deuterium bottleneck at $T_{\text{D-bottle}} \approx 0.07$ MeV	D-via-I	§3.3. Photodissocia- tion/formation rate competition + η_b inheritance.
11	Helium-4 mass fraction $Y_p \approx 0.247$ via neutron-capture chain	D-via-I	§3.4. Composition of rows 8-10 + standard nuclear-reaction chain.
12	D/H, ${}^3\text{He}/\text{H}$, ${}^7\text{Li}/\text{H}$ abundance predictions via standard BBN codes	I	§3.5. Inherits standard BBN-code outputs.
13	Substrate-parameter Θ_{ED} sensitivity: $dY_p/d \ln \Theta_{\text{ED}} \approx 0.05$ at observed value	D-via-I	§3.6. Composition of Route A4 + standard BBN H -sensitivity.
14	Substrate-c bound preserved throughout BBN epoch	D-via-I	§3.1. Paper_012 + Paper_089.
15	Baryon-to-photon ratio $\eta_b \approx 6.1 \times 10^{-10}$ inherited from CMB	I	Sub-type per Paper_095 §3.2.1: <i>IC-INHERITED</i> . §3.3. Planck-derived η_b .

#	Step	Label	Notes
16	Early-universe substrate state (post-inflation reheating; particle content) inherited from Cos_01 + standard cosmology	I	Sub-type per Paper_095 §3.2.1: <i>IC-INHERITED</i> . §2.IC. Cos_01 post-BBB-ignition framework + standard cosmology for reheating-to-BBN evolution.
17	Lithium-7 puzzle (observed below standard prediction): inherited unresolved standard-BBN puzzle	OPEN (non-load-bearing)	§3.5. Independent of substrate-ontology framing; standard-BBN puzzle.
18	Verdict M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime	A→position	Per Paper_095 §3.3 line 77. Five-anchor verdict-sync: status / abstract / §1 / this row / §6 all M3.

Audit summary: zero pure-D rows; **eight substantive D-via-I rows** (6, 8-11, 13, 14); five I rows (2, 3, 4, 5, 12); one substrate-parameter-INHERITED row (7); two IC-INHERITED rows (15, 16); one non-load-bearing OPEN (17 — lithium puzzle); one P row (1); one A→position verdict row (18). **No load-bearing OPEN items.** Upstream Dyn_02 M2 (RDE chain-class identification pending Route C1) is inherited at upstream status without load-bearing on Cos_02's net M3 verdict (Cos_02 inherits continuum-Friedmann at standard-cosmology level via DCGT). No new substrate primitives; no paper-specific postulates.

§5 Falsification Criteria

- **F1 (sharpest):** Observed primordial He-4 mass fraction Y_p or deuterium D/H ratio significantly inconsistent with substrate-graph-derived $H(t)$ at RDE composed with standard nuclear physics — beyond observational + standard-BBN-uncertainty (~5% precision). Refutes §3.1-§3.4 + audit rows 6-11. Currently consistent: $Y_p \approx 0.245\text{-}0.255$, $D/H \approx (2.5\text{-}2.6) \times 10^{-5}$, CMB-BBN η_b cross-consistency at ~5% precision.
- **F2:** Observed deuterium D/H or He-3 abundance significantly outside standard-BBN prediction range under substrate-parameter Θ_{ED} scaling. Refutes §3.6 + audit row 13.
- **F3:** Observed BBN-epoch freeze-out temperature signatures (extractable from precision Y_p + D/H + ${}^7\text{Li}$ joint analyses) significantly inconsistent with ED $H(t)$ scaling. Refutes §3.2-§3.3 + audit rows 8-10.
- **F4 (substrate-research-frontier-conditional):** Discovery of substrate-parameter Θ_{ED} value from independent substrate-research-frontier work (RA-OPEN-4a closure) inconsistent

with the value required to reproduce observed H_0 and BBN abundances simultaneously. **Empirical refutation not currently available** (depends on RA-OPEN-4a closure). Refutes §3.6 + audit row 7 (substrate-parameter-INHERITED level).

§6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton interpretation of QM; Wolfram Ruliad / hypergraph-rewriting; causal-set program, Sorkin et al.), not within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

BBN in ED is the substrate-graph reading of standard light-element nucleosynthesis: freeze-out competition between standard nuclear-reaction rates Γ_{nuc} and the substrate-graph-derived expansion rate $H(t)$ during the radiation-dominated era. The substrate-graph composition supplies the cosmological $H(t)$ at standard-cosmology level via DCGT continuum bridge (Cos_01 + Cos_05 cosmological-expansion framework + Route A4 substrate-parameter-INHERITED H_0 scaling at Paper_027-analog tier); standard nuclear physics + standard cosmology + standard BBN-code calculations are inherited at textbook level. Substrate-c bound preserved throughout BBN epoch. Substrate-ontology distinguishing content vs Λ CDM standard BBN: substrate-parameter Θ_{ED} scaling channel for BBN abundances via Route A4 inheritance (Cos_02 §3.6); same standard Y_p , D/H, ^7Li predictions inherit standard BBN-code outputs but now carry explicit substrate-parameter scaling.

Verdict: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime hydrodynamic-window per Paper_095. Eight D-via-I rows; five I rows; one substrate-parameter-INHERITED row (Route A4); two IC-INHERITED rows (Cos_01 post-inflation reheating + CMB η_b); zero pure-D rows; no paper-specific postulates; no new substrate primitives; **no load-bearing OPEN items**. One non-load-bearing OPEN (lithium-7 puzzle; inherited standard-BBN puzzle, independent of substrate-ontology framing). Upstream Dyn_02 M2 (RDE chain-class identification pending Route C1) inherited at upstream status as substrate-research-frontier tightening per Preamble item 7; not load-bearing for Cos_02's net M3 via continuum-Friedmann inheritance. **Empirical-anchor strength:** BBN is one of the most stable empirical anchors in cosmology — 50+ years of observational success at ~5% precision across $Y_p + \text{D/H} + \eta_b^{\text{CMB}}$ vs η_b^{BBN} cross-consistency, exceeding the empirical-anchor strength of even the strongest GW postdictions in the corpus (Hulse-Taylor 0.1%-precision orbital decay; NANOGrav 2023 nHz signal). The substrate-parameter Θ_{ED} sensitivity channel via Route A4 is the genuine ED-distinguishing content; future precision improvements in D/H + Y_p observations directly refine this sensitivity.

Cross-arc consequences. Upstream, Cos_02 inherits unchanged from Cos_01 (M3), Cos_05 (M3), Dyn_02 (M2 — chain-class OPENs not aggravated per Preamble item 7); no other arc impact. Downstream, Cos_02 *enables* Paper_ED_Cos_03 (CMB acoustic) via composition with Cos_01 + Cos_05 + standard recombination physics, and Paper_ED_Cos_04 (linear structure formation) via Cos_02 baryon-fraction content + Cos_05 expansion history; *strengthens* the SCBU BH-horizon + cosmic-decoupling projections via cross-arc cosmological-history consistency between BBN-epoch + recombination-epoch + late-LDE asymptote at substrate-parameter-INHERITED level via Route A4; *strengthens* the empirical-prediction sector via the $\Omega_b h^2$ BBN-derived vs CMB-derived cross-consistency check now carrying explicit substrate-graph framework. Whether each

downstream paper closes at M3 depends on its own substrate-graph chain; not forecast here.

End Paper_ED_Cos_02.